



西北工业大学
NORTHWESTERN POLYTECHNICAL UNIVERSITY

Humanoid Control

Locomotion

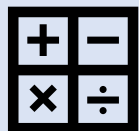
Teleoperation

Motion Tracking

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Major Concern



Model-based: MPC, LQR, ...

- Strong interpretability & safety
- Data efficiency
- Suitable for structured problems
- Modeling difficulty
- Struggle with real-world uncertainties
- High computation payload at inference



Learning-based: RL, IL, E2E, ...

- Offer robustness to uncertainties
- Less modeling effort but complex reward design
- Strong perception integration & multi-modal
- Data hungry
- Poor interpretability
- High training cost of computation and time

Pros (+)
Cons (-)

Respective Concern

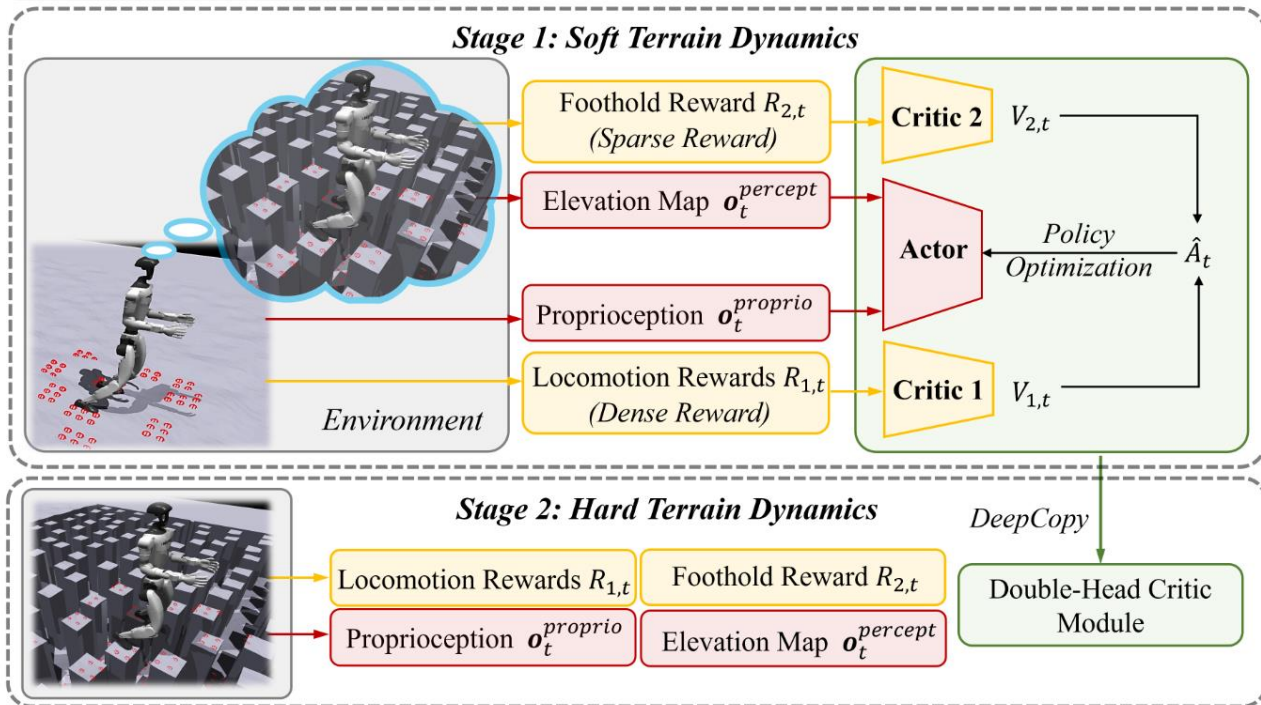
- How to achieve agile and stable locomotion in reality? ——— BeamDojo, Attn-Based Map & ASAP
- How to produce human-like and versatile motions? ——— BeyondMimic, Homie & KungfuBot
- How to scale humanoid control? ——— SONIC

Question ——— Solution

**How to achieve agile and stable
locomotion in reality?**

BeamDojo: Learning Agile Humanoid Locomotion on Sparse Footholds

(a) Training in Simulation

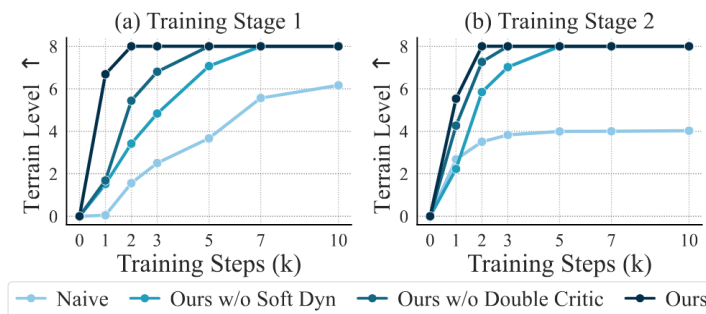


Double Critic

$$\hat{A}_t = w_1 \cdot \frac{\hat{A}_{1,t} - \mu_{\hat{A}_{1,t}}}{\sigma_{\hat{A}_{1,t}}} + w_2 \cdot \frac{\hat{A}_{2,t} - \mu_{\hat{A}_{2,t}}}{\sigma_{\hat{A}_{2,t}}}$$

Two-stage RL Framework

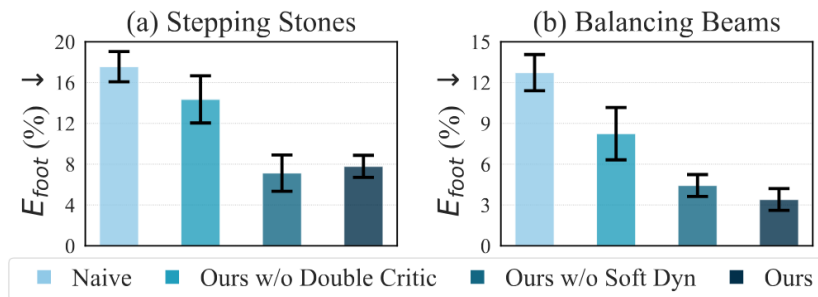
Encourage sufficient trial-and-error exploration from flat terrain to actual terrain



Ablation Study

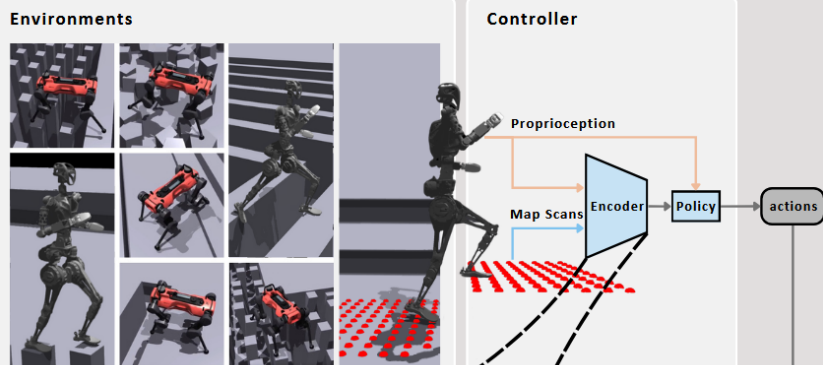
Sampling-based Foothold Reward

$$r_{\text{foothold}} = - \sum_{i=1}^2 \mathbb{C}_i \sum_{j=1}^n \mathbb{1}\{d_{ij} < \epsilon\}$$

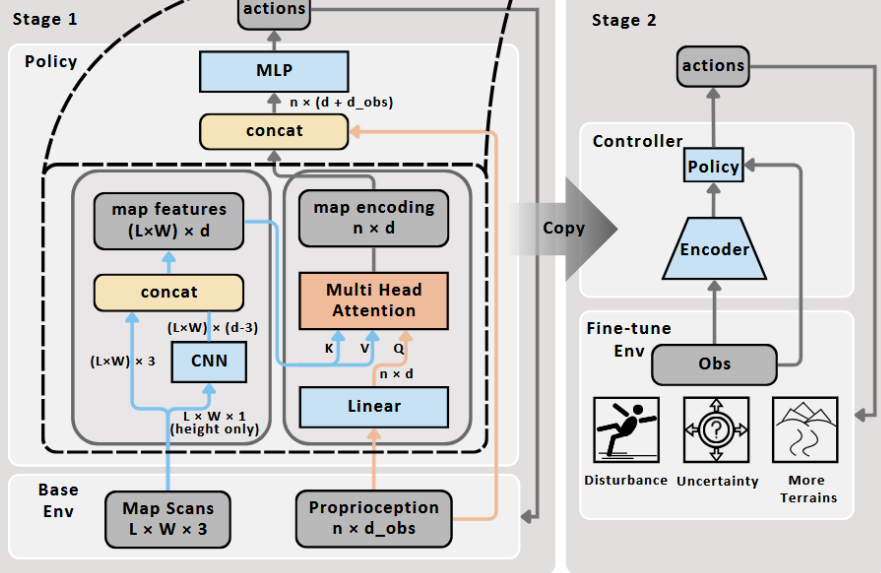


Attention-Based Map Encoding for Learning Generalized Legged Locomotion

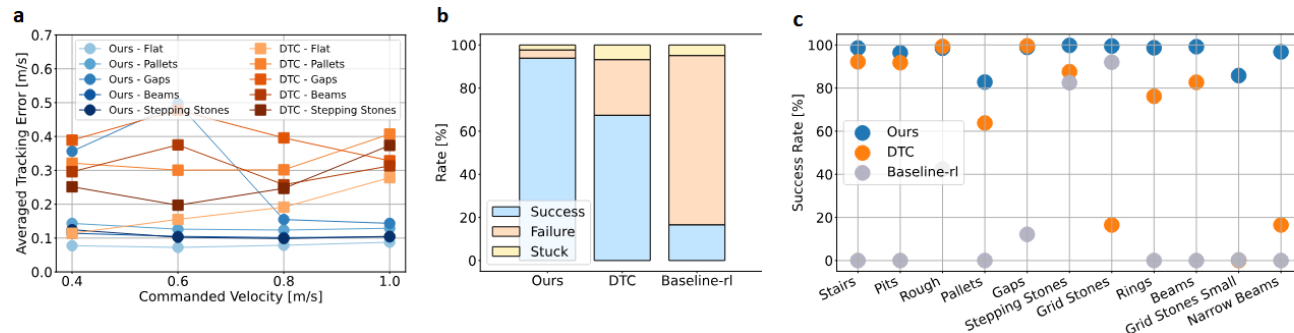
A Locomotion strategies abstract



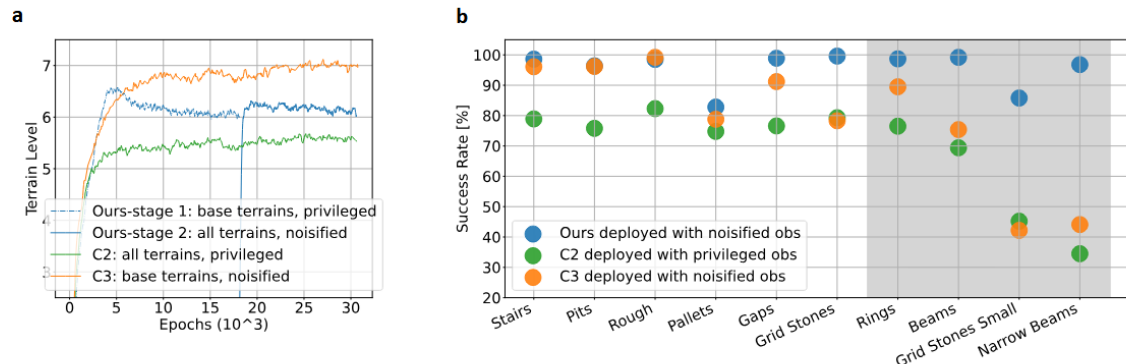
B



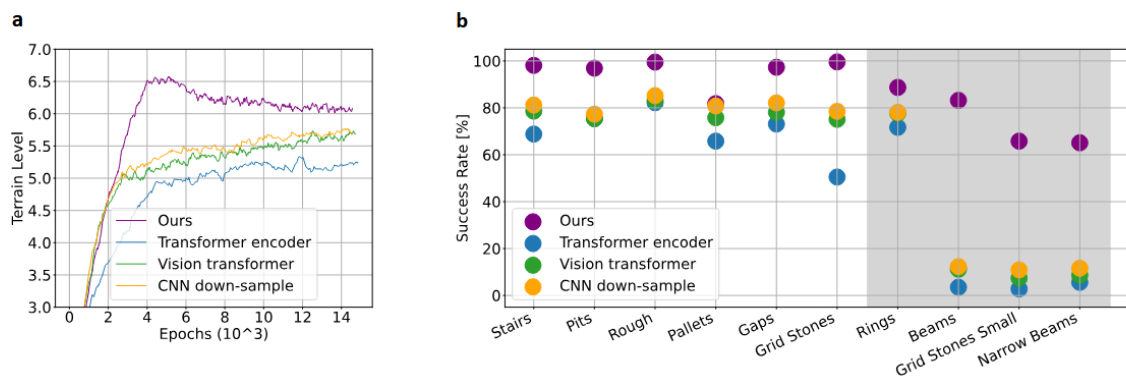
A. Benchmark with DTC and Baseline RL

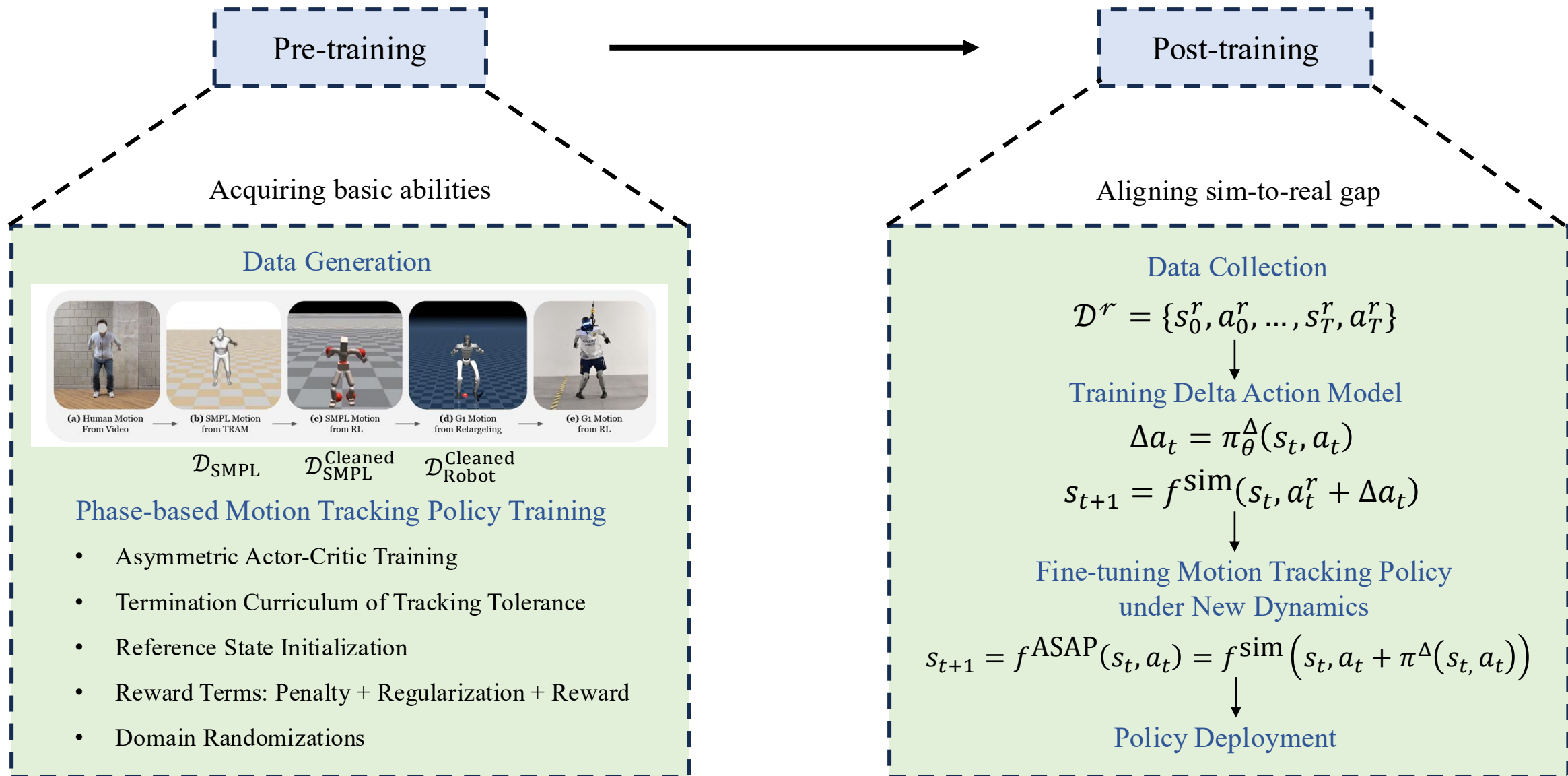


B. Ablation study on two-stage training

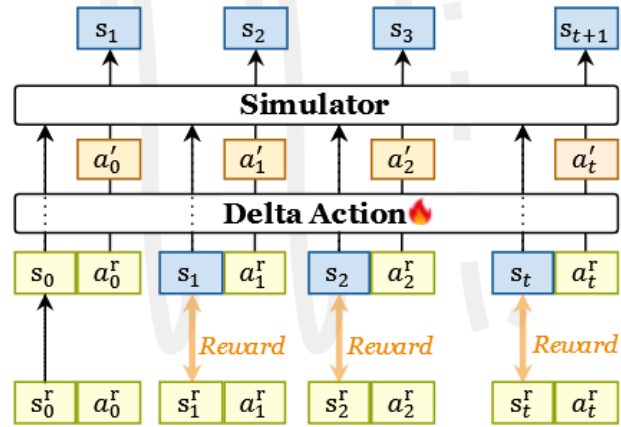
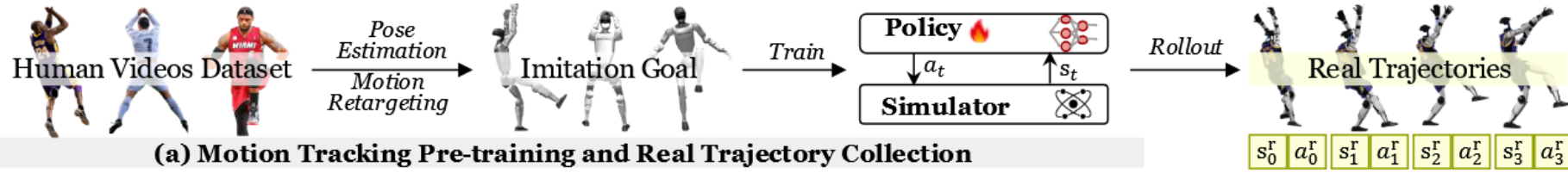


C. Ablation study on network structure

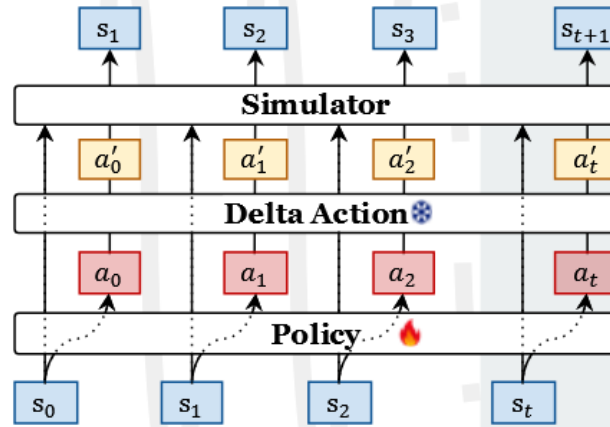




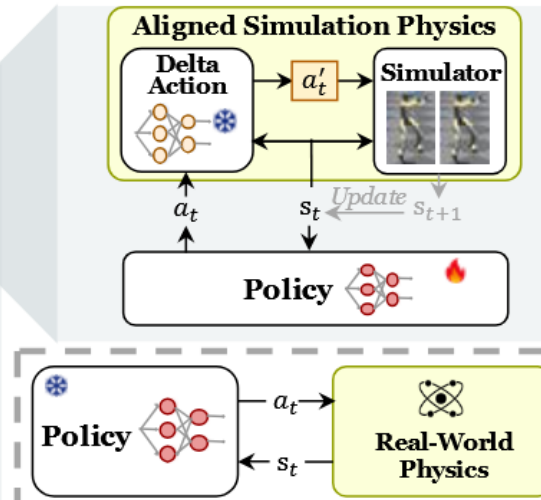
Overview of ASAP



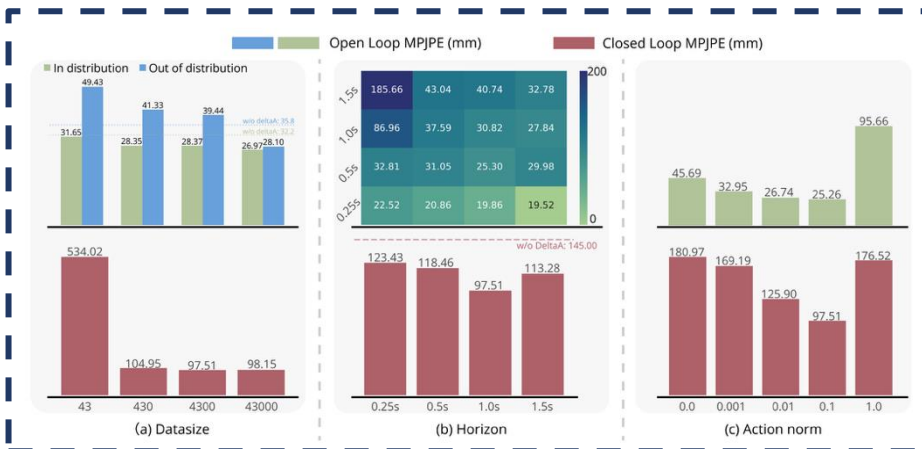
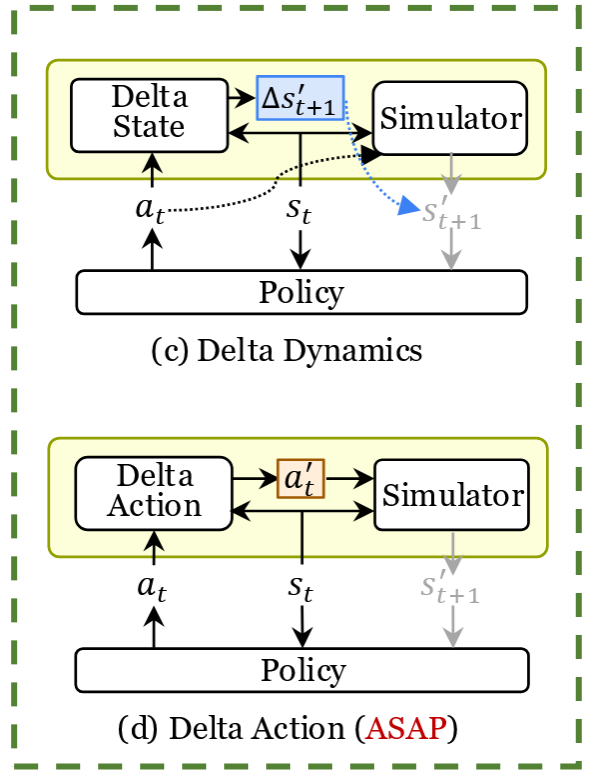
(b) Delta Action Model Training



(c) Policy Fine-tuning



(d) Real World Deployment



Analysis on key factors

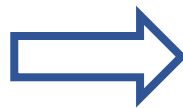
- Open-loop testing measures one-step prediction accuracy
- Closed-loop testing evaluates long-term stability and robustness under real-world execution

Advantages of action correction vs. state correction

- Physical consistency
- Less parameters
- Simulator serves as low pass filter

**How to produce human-like and
versatile motions?**

- **Heavy** domain randomization
- **Heavy** reward regularization
- **Complex** observations



- same model parameters, reward function, and hyperparameters across a diverse set of skills and different physics engines

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_{\varepsilon}(\mathbf{z}|\mathbf{x})} [\|\mathbf{x} - \mathcal{D}(\mathbf{z})\|^2] + \beta D_{\text{KL}}(q_{\varepsilon}(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$$

$$\mathcal{L}_{\text{Diffusion}} = \mathbb{E} [\|z_{\phi}(\mathbf{z}^k, k) - \mathbf{z}^0\|^2]$$

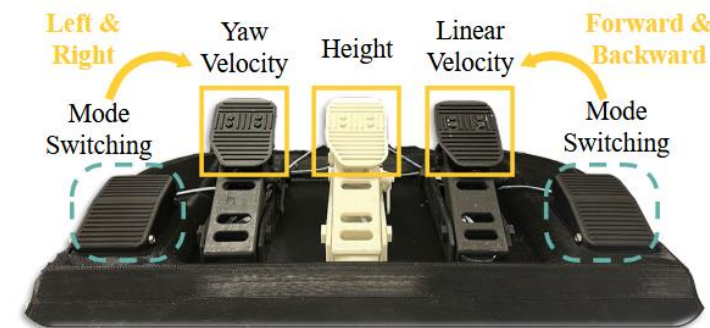
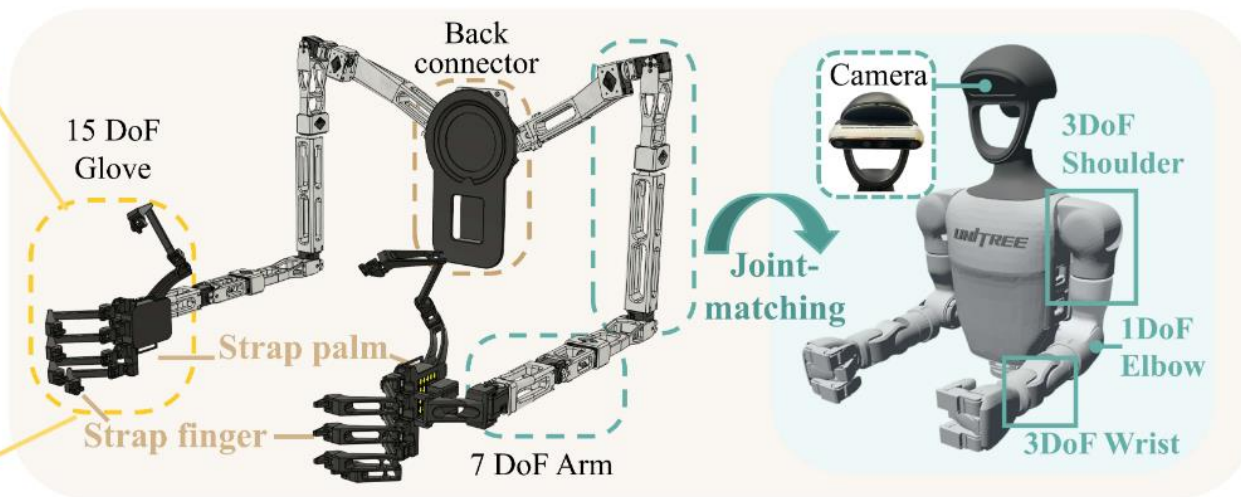
$$\mathbf{z}^{k-1} = \alpha_k \left(\mathbf{z}^k - \gamma_k \left(\mathbf{z}^k - z_{\phi}(\mathbf{z}^k, k) \right) \right) + \sigma_k \mathcal{N}(\mathbf{0}, \mathbf{I})$$

HOMIE: Humanoid Loco-Manipulation with Isomorphic Exoskeleton Cockpit

(a) Motion-sensing Gloves



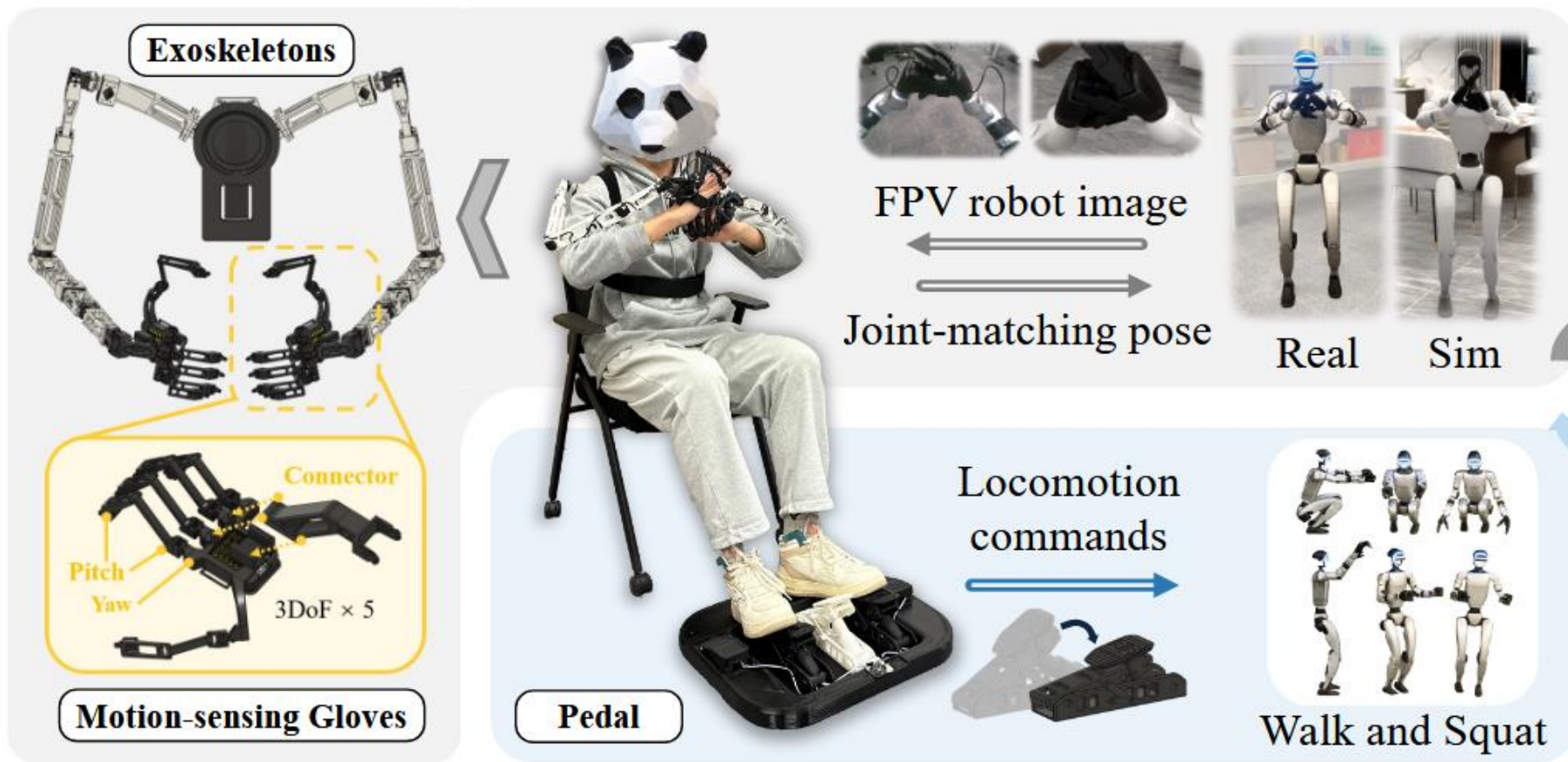
(b) Matching Between Isomorphic Exoskeletons and Humanoids



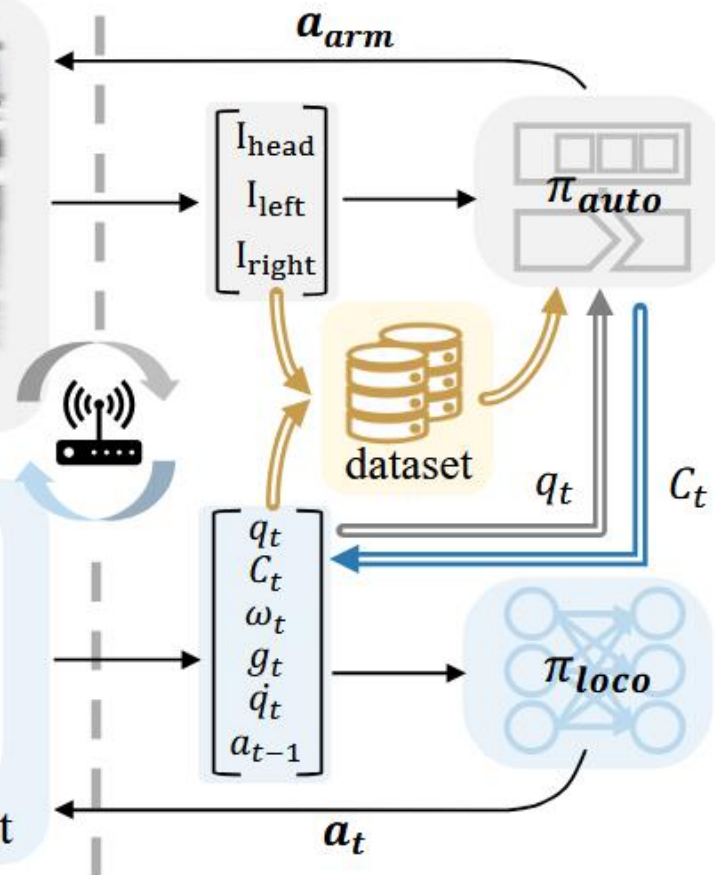
Hardware of Semi-autonomous Teleoperation System

- Gloves utilizing Hall sensors: freely adapt to **any model** of dexterous hands
- Exoskeleton: delivers **faster and more precise** arm control by eliminating the reliance on inverse dynamics
- The pedal design: **removes the need for continuous synchronized walking** between the operator and the robot

(a) Humanoid Whole-body Teleoperation



(b) Policy

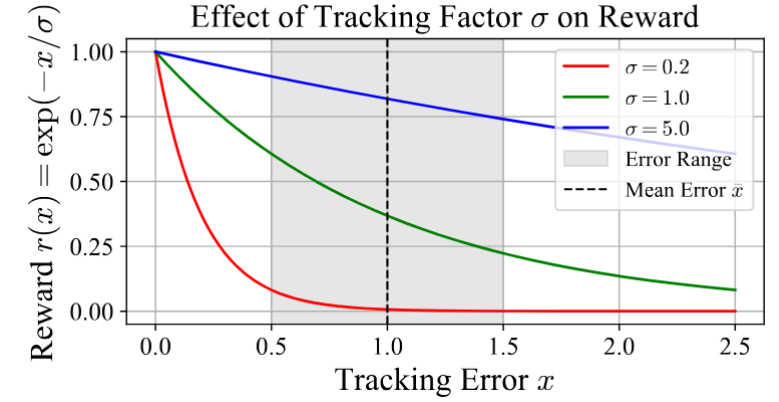
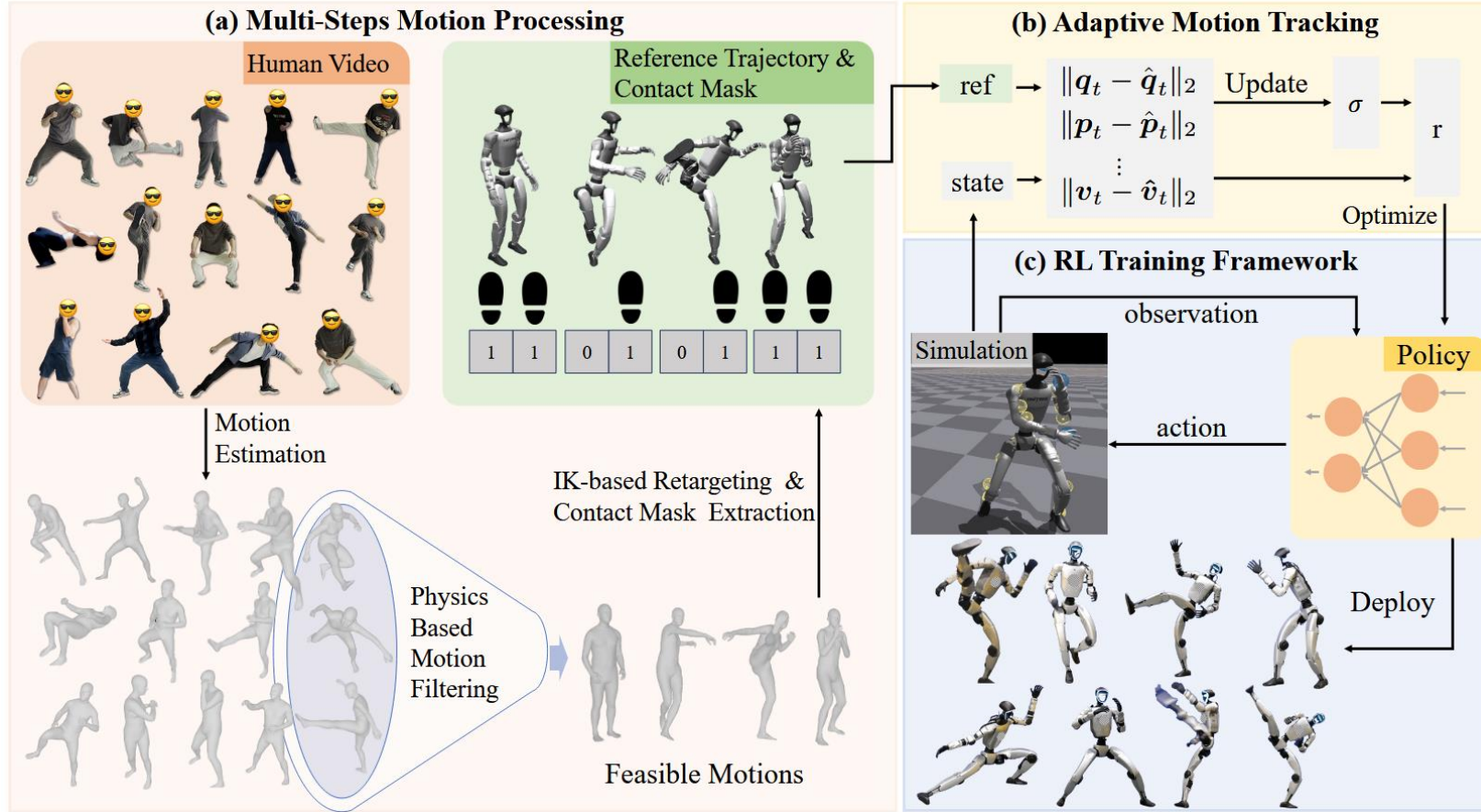


- Upper-body Pose Curriculum for training π_{loco}

- Height Tracking Reward:
$$r_{knee} = - \left\| (h_{r,t} - h_t) \times \left(\frac{q_{knee,t} - q_{knee,min}}{q_{knee,max} - q_{knee,min}} - \frac{1}{2} \right) \right\|$$

- Symmetry Utilization

KungfuBot: **Physics-Based** Humanoid Whole-Body Control for Learning **Highly-Dynamic** Skills



$$J^{\text{in}}(x, \sigma) = \sum_{i=1}^N \exp(-x_i/\sigma)$$

$$\max_{x \in \mathbb{R}_+^N} J^{\text{in}}(x, \sigma) + R(x)$$

$$\max_{\sigma \in \mathbb{R}_+} J^{\text{ex}}(x^*),$$

$$\text{s.t. } x^* \in \arg \max_{x \in \mathbb{R}_+^N} J^{\text{in}}(x, \sigma) + R(x).$$

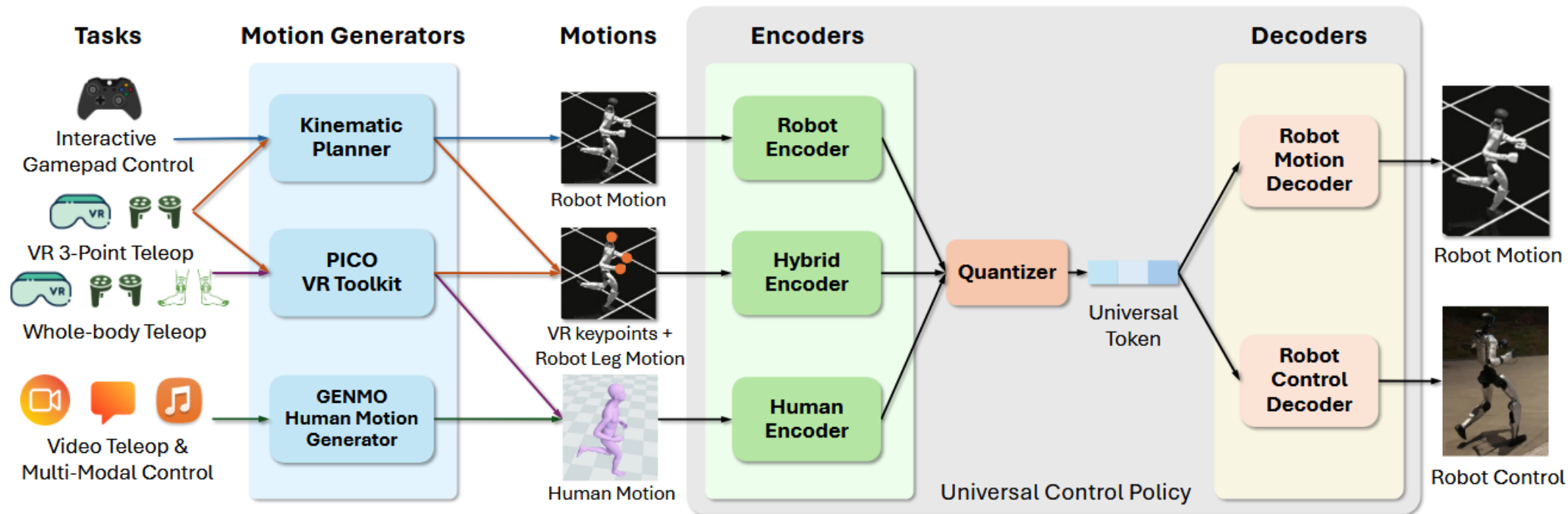
$$\Delta d_t = \|\overline{p_t^{\text{CoM}}} - \overline{p_t^{\text{CoP}}}\|_2 < \epsilon_{\text{stab}}$$

- Boundry stability
- Maximum instability gap < threshold

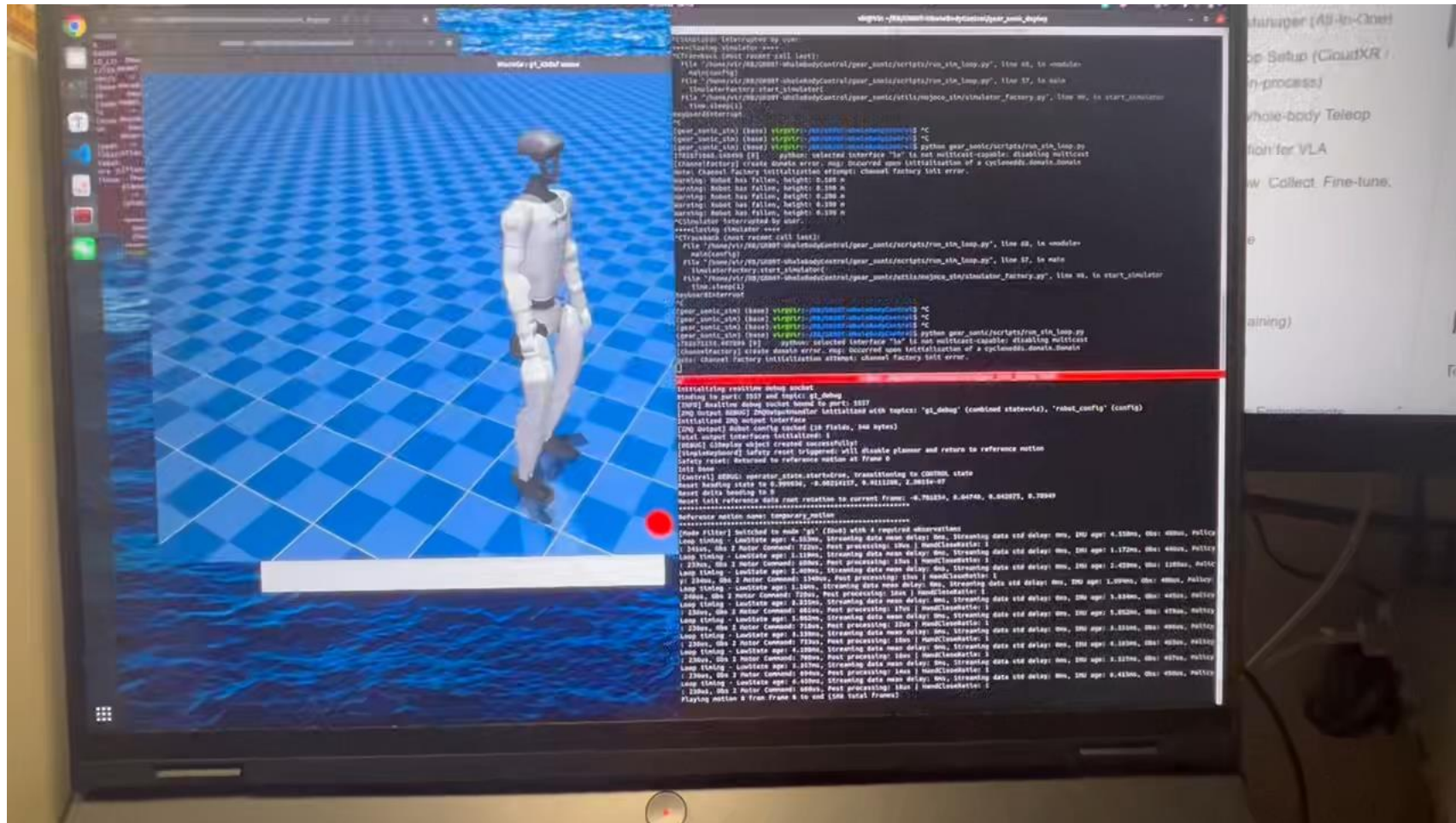
$$\sigma^* = \left(\sum_{i=1}^N x_i^* \right) / N.$$

How to scale humanoid control?

SONIC: Supersizing Motion Tracking for **Natural** Humanoid Whole-Body Control



$$\begin{aligned}
 L &= \mathcal{L}_{ppo} + \mathcal{L}_{recon} + \mathcal{L}_{token} + \mathcal{L}_{cycle} \\
 \mathcal{L}_{recon} &= \|\mathcal{D}_r(z_r) - \mathbf{g}_r\|^2 + \|\mathcal{D}_r(z_h) - \mathbf{g}_r\|^2 + \|\mathcal{D}_r(z_m) - \mathbf{g}_r\|^2 \\
 \mathcal{L}_{token} &= \|z_r - z_h\|^2 + \|z_r - z_m\|^2 + \|z_m - z_h\|^2 \\
 \mathcal{L}_{cycle} &= \|\mathcal{E}_r(\mathcal{D}_r(z_h)) - z_r\|^2
 \end{aligned}$$



SONIC

Modify src and scripts before reproducing



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Thanks for listening!

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